

VŨ ĐỨC DUY

Full Stack Developer

so7ngovanso@gmail.com 0973628663 Ha Noi, Viet Nam <https://www.linkedin.com/in/duy-vu-5bb6291a4/> github.com/vana

SUMMARY

3+ years building scalable web applications. Passionate about clean architecture, developer experience, and shipping products users love.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, SQL
Frontend	React, Tailwind CSS
Backend	Node.js, Express, NestJS, Flask, Fastapi, Hapi.js
Databases	PostgreSQL, MongoDB, Redis, Prisma
DevOps & Tools	Docker, AWS, GitHub Actions, Nginx

EXPERIENCE

Full Stack Engineer

Mar 2025 – Present

iArt Holdings

- Replaced full-table-scan BullMQ job with a fan-out scheduler (400 orders/tick, 12-order chunks), eliminating DB memory spikes and reducing unit creation latency from minutes to seconds.
- Designed outbound webhook system with Redis-based deduplication (TTL 7 days), rate limiting (150 req/min), and 8-attempt exponential backoff — ensuring zero duplicate deliveries to vendors.
- Resolved race condition between order cancellation and unit creation using Redis distributed locks (SET NX EX) and a re-fetch with status guard before DB writes.
- Built timezone-aware production dashboard with 6 KPI blocks, SLA breach alerts, and configurable date ranges — serving warehouse teams across Vietnam and London timezones.
- Implemented a Gap Engine calculating working vs. idle time on-read from MongoDB activity logs (10-minute threshold, per-user per-stage), eliminating scheduled pre-aggregation jobs.
- Diagnosed a PostgreSQL Nested Loop Anti-Join issue (~100k probe iterations) on "orders without units" query; applied raw SQL + LEFT JOIN as a mitigation.
- Set up CI/CD pipeline with GitHub Actions + HashiCorp Vault for runtime secret injection, Docker multi-stage builds, and automated deployment to AWS EC2 via ECR.
- Architected a polyglot data layer: PostgreSQL (101 models), MongoDB for high-volume activity logs, Redis for queues, distributed locks, and deduplication cache.

PodFactory — Production Management System

Stack: Node.js, Hapi.js, TypeScript, React 18, PostgreSQL, MongoDB, Redis, BullMQ, Socket.IO, Docker, AWS, Vault

- Built full production workflow system: Pick → Print → QC → Pack → Ship for a DTG garment factory
- Replaced full-table-scan BullMQ job with fan-out scheduler (400 orders/tick), cutting unit creation latency from minutes to seconds
- Designed outbound vendor webhook system with Redis deduplication, rate limiting (150 req/min), and 8-attempt exponential backoff
- Resolved order cancellation race condition using Redis distributed locks (SET NX EX) with status re-fetch guard
- Built timezone-aware dashboard with 6 KPI blocks, SLA breach alerts, and 7/14/30-day trend charts
- Implemented Gap Engine calculating working vs. idle time on-read from MongoDB activity logs, no pre-aggregation needed
- Diagnosed PostgreSQL Nested Loop Anti-Join (~100k iterations) on orders query; mitigated with raw SQL + LEFT JOIN
- Set up CI/CD: GitHub Actions + HashiCorp Vault + Docker multi-stage build → AWS ECR/EC2 automated deploy

PodFactory PDA App

Stack: Flutter, Dart, REST API, Barcode Scanner (ZXing)

- Warehouse management PDA app for DTG print fulfillment
- Covers full WMS workflow: Inbound, Put Away, Replenishment, Cycle Count, Picking, Shipping
- Barcode/QR scan integration for inventory operations
- Cross-platform: Android (primary), Windows

iArt Pod Client (DTG Print Factory Tool)

Stack: C# .NET 8, WinForms, Python, AutoIt3

- Automated DTG/DTF print job pipeline with Python layout tools
- Integrated Brother GTX + ONYX RIP via AutoIt3 automation
- Reduced fabric waste with roll merge optimizer

Walmart Product Crawler

Stack: Python, FastAPI, Selenium, MongoDB, Docker

- Crawled 100k+ products from Walmart.com with automated browser automation
- Bypassed anti-bot detection using proxy rotation and undetected-chromedriver
- Built REST API for querying and filtering crawled product data
- Deployed with Docker, managed multi-profile browser sessions via MultiloginX

Back End Developer

Mar 2023 – Oct 2024

Jeeves

- Designed and implemented robust RESTful APIs to manage corporate profiles, business registration documents, and credit approval workflows.
- Engineered a strict Role-Based Access Control (RBAC) system using JWT/OAuth2, ensuring secure data isolation and restricting approval actions exclusively to Managers while granting view-only access to Support staff.
- Designed and optimized scalable database schemas to efficiently track real-time application states and audit history.
- Developed secure file upload and storage modules to handle sensitive business licenses and legal documents compliance-wise.

Internal Admin Portal & Underwriting APIs

Stack: Node.js, Express.js, PostgreSQL, Redis, JWT, OAuth2, Docker

- Developed robust RESTful APIs to efficiently handle corporate profiles, business registration documents, and multi-stage credit approval workflows (CRUD).
- Implemented a strict Role-Based Access Control (RBAC) system integrated with JWT/OAuth2 to secure sensitive corporate data and isolate permissions.
- Designed and optimized relational/non-relational database schemas to track real-time application states and maintain system audit logs.
- Enforced strict backend authorization, restricting critical approval actions exclusively to Managers while granting view-only access to Support staff.
- Built secure file upload and handling modules for processing sensitive legal documents and business licenses.

Back End Developer

Apr 2021 – Oct 2022

FPT Software

- Contributed to the development of FPT's next-generation AI conversational platform.
- Optimized real-time data processing pipelines for high-concurrency AI services.

Real-time AI Voice Assistant Platform

Stack: Python (FastAPI), LangChain, OpenAI API, WebSocket, Redis, Docker

- Designed and implemented scalable backend systems using Python (FastAPI) to support real-time AI interactions
- Integrated Large Language Models (LLMs) like OpenAI and Llama via LangChain to enhance chatbot reasoning and response quality
- Engineered real-time voice processing pipelines (STT/TTS) using WebSockets for seamless, low-latency communication
- Optimized end-to-end system latency, reducing AI response wait time by 40% through efficient prompt engineering and caching

PROJECTS

Diabetes Prediction — ML Training & Model Serving Pipeline — GitHub

Stack: Python, FastAPI, scikit-learn, XGBoost, LightGBM, MLflow, Docker, Prometheus, OpenTelemetry (Jaeger), Locust, Pydantic

Built a training pipeline comparing 10 ML algorithms (XGBoost, LightGBM, SVM, Random Forest, MLP, AdaBoost, Naive Bayes, Decision Tree, KNN, Logistic Regression) for a diabetes classification task, evaluated with accuracy, precision, recall, F1, and ROC-AUC. Deployed MLflow (containerized with Docker) for experiment tracking, logging params/metrics/artifacts, and managing a versioned model registry. Built a REST API for model serving using FastAPI and Pydantic validation, supporting concurrent serving of multiple models loaded from the MLflow registry. Instrumented the serving system with observability: distributed tracing via OpenTelemetry/Jaeger (measuring per-step latency: dataframe conversion → scaling → prediction) and metrics export via Prometheus. Wrote load tests with Locust simulating real traffic using 200+ sample requests to evaluate API throughput and latency under load. Wrote unit tests (pytest) to validate model prediction accuracy on real data.

EDUCATION

Electric Power University

Sep 2020 – Aug 2025

Engineer's Degree, Software Engineering

GPA: 3.2/4.0